

A/B Test Analysis

Measuring the Impact of a New Webpage Design on Conversion Rate

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Extended analysis — based on MSc AI Project 3: Statistics for Data Analysis

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The Business Question

An e-commerce company redesigned its webpage and wanted to know whether the new design genuinely improved customer conversion — or whether any observed difference could simply be due to chance.



Control Group

Visitors continued to see the existing webpage design.



Treatment Group

Visitors were shown the newly redesigned webpage.

The question:

Did the treatment page perform better by an amount that is unlikely to have occurred by chance?

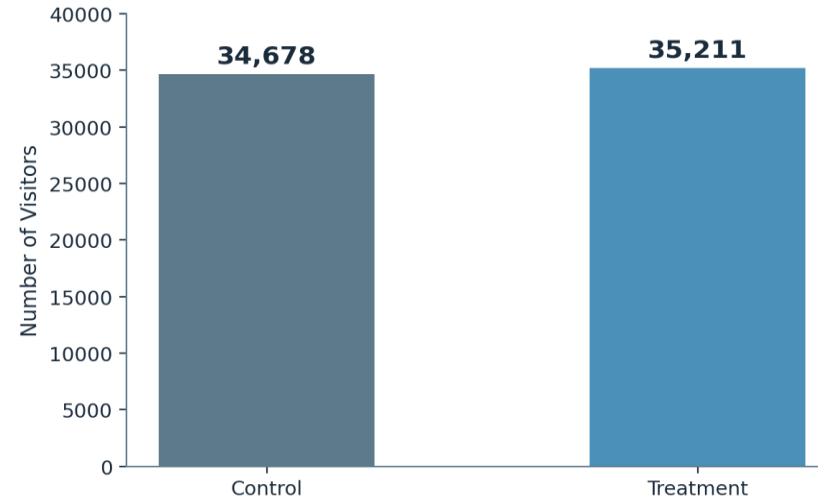


The Data

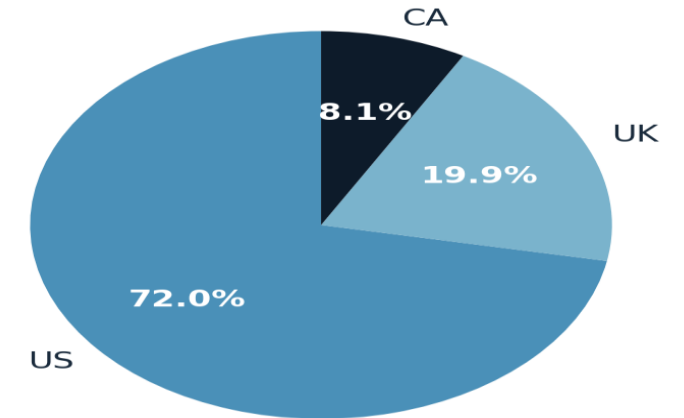
69,889

Total visitor records analysed

Sample Size by Group (n = 69,889)



Visitor Distribution by Country



Dataset structure

- country — visitor's country of origin (US, UK, CA)
- group — experimental assignment (control or treatment)
- converted — conversion outcome (0 = no, 1 = yes)

No missing values were identified during data validation. Group sizes were close to balanced, supporting a direct comparison between the two experimental arms.



Methodology: A Simulation-Based Hypothesis Test

H₀ The treatment page performs no better than the control page.

H₁ The treatment page performs better than the control page.

1

Shared null rate

Assume both groups share one true conversion rate — the overall observed rate across the full dataset.

2

500 simulations

Draw 500 simulated samples per group, matching original sample sizes, using the shared null rate.

3

Null distribution

Record the simulated treatment-control difference each time, building a distribution of expected differences under “no effect”.

4

Compare & decide

Compare the actual observed difference to this distribution to calculate a p-value.

Results

10.53%

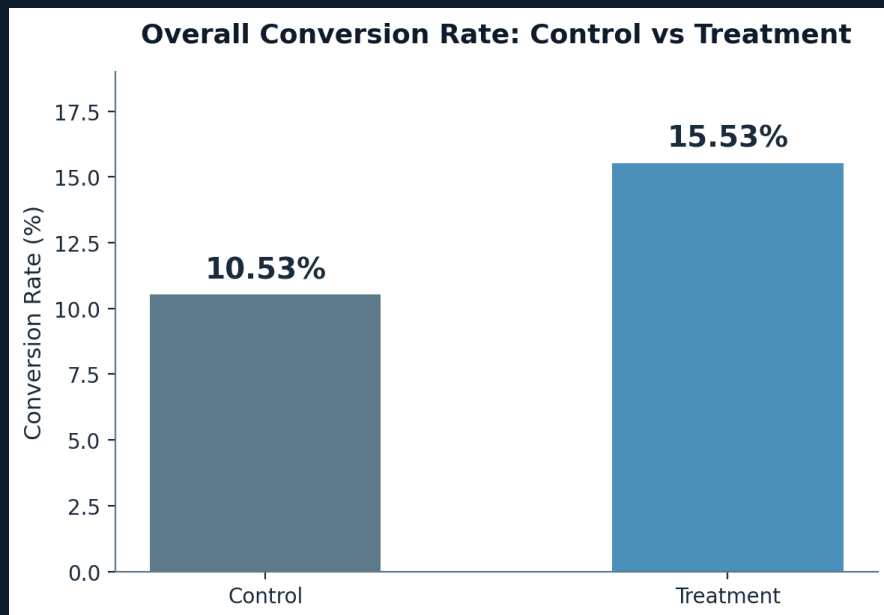
Control conversion rate

15.53%

Treatment conversion rate

+47.4%

Relative improvement



p-value < 0.002

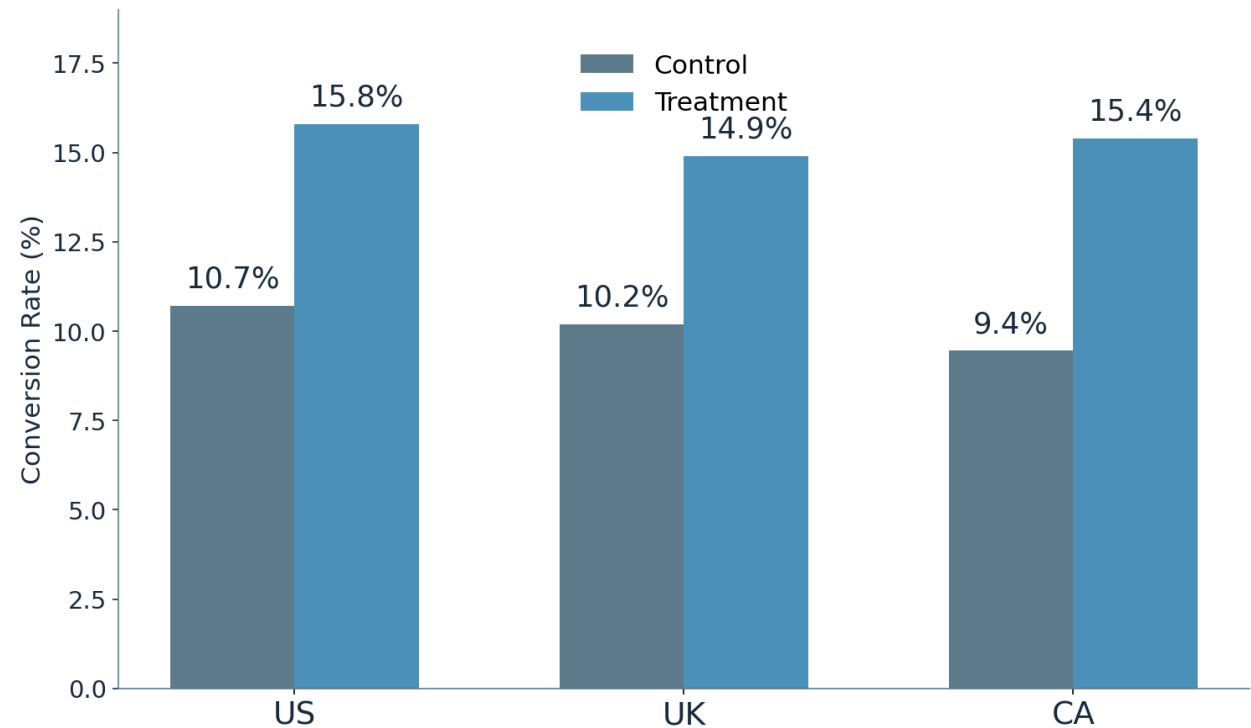
The observed difference of 5.01 percentage points was substantially larger than any difference produced under the simulated null distribution, and well below the $\alpha = 0.05$ significance threshold.

Decision: the null hypothesis is rejected.



The Effect Holds Across All Three Markets

Conversion Rate by Country and Group



Conversion rate by country and group

Group	US	UK	CA
Control	10.7%	10.2%	9.45%
Treatment	15.8%	14.9%	15.4%

The magnitude of improvement was broadly consistent across markets, with absolute gains ranging from approximately 4.7 to 6.0 percentage points.

This consistency suggests the effect of the redesign is general rather than driven by behaviour specific to a single market.



Decision and Recommendation



Deploy the treatment page — the improvement is statistically significant and consistent across all three markets studied.

Recommended next steps

- Deploy the treatment page to all users across the markets included in this experiment
- Monitor post-deployment conversion rate to confirm the improvement is sustained
- Run follow-up experiments to isolate which design elements drove the increase
- Extend future testing to additional regions and customer segments

Business impact

A relative improvement of 47.4% in conversion rate, if sustained at scale, represents a material opportunity for increased customer acquisition and revenue.

The consistency of the result across three markets increases confidence that the effect reflects a genuine improvement in page design — not a market-specific anomaly.



Limitations

1

The analysis covers only visitors from the US, UK, and Canada; results may not generalise to other markets.

2

Conversion rate was the sole performance metric evaluated — other outcomes such as average order value were not assessed.

3

No demographic, behavioural, or device-level data was available, limiting segment-level insight.

4

Long-term customer retention and lifetime value were not measured.

5

External factors influencing visitor behaviour during the experiment were not controlled for.



Observation vs Evidence

An observed improvement in conversion rate can always be seen in a sample. The question that matters for decision-making is whether that improvement is large enough, and consistent enough, to be considered reliable rather than the product of random variation.

The simulation-based hypothesis test applied here translates a single observed difference into a statement about how likely that difference would be if no real effect existed — quantifying uncertainty rather than eliminating it, and allowing a business to act with a stated, defensible level of confidence.

Skills demonstrated

- A/B Testing
- Statistical Analysis
- Hypothesis Testing
- Probability
- Data Interpretation
- Python — Pandas, NumPy
- Business Analytics